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Reb-Tech_Submission_04_Aakash (Notes)

1. Descriptive Statistics: Theoretical Foundation

1.1 What is descriptive statistics?

Descriptive Statistics is the branch of statistics dedicated to summarizing, organizing, and presenting data in a meaningful way. Unlike inferential statistics, which uses sample data to make predictions or draw conclusions about a broader population, descriptive statistics focuses solely on describing the features of the observed dataset at hand.

Suppose our dataset contains thousands of payment transactions. Looking at 50,000 individual transaction records doesn't immediately tell us:

- What is the typical transaction amount?
- How much do transaction amounts vary?
- Are most transactions small or large?
- Is the data symmetric or skewed?
- Are there unusually large transactions?
- Where does the middle 50% of the data lie?

Descriptive statistics compress a large dataset into a small number of meaningful statistical measures.

1.1.1 Measures of Central Tendency

Measures of central tendency locate the single central value that best represents an entire distribution of quantitative data.

a) Mean (Arithmetic Average)

The arithmetic mean is the sum of all observations divided by the number of observations.

$$\bar{x} = \frac{1}{N} \times \sum_{i=1}^N x_i$$

It includes every data point, making it mathematically tractable for further statistical modeling. It is highly sensitive to extreme values (outliers) and skewed distributions.

b) Median (Middle Position)

The median is the exact middle point of an ordered dataset (or the average of the middle two values when N is even). Robust to outliers and extreme skewness. It accurately represents the center of non-symmetric distributions.

c) Mode (Most Frequent)

The mode is the value or category that appears most frequently in a variable distribution. Applicable to both numerical and categorical data.

The relationship between mean and median can tell us something about the shape of a distribution.

i. Approximately symmetric distribution

$$\text{mean} \approx \text{median}$$

ii. Right-skewed distribution

$$\text{mean} > \text{median}$$

Mean > median is a useful indication of right skewness, but it is not a formal proof by itself. Later, our histograms and box plots will help us investigate the distribution properly.

iii. Left-skewed distribution

$$\text{mean} < \text{median}$$

1.1.2 Measures of Dispersion and Variability

Measures of dispersion quantify how spread out the observations are around the center of the distribution.

a) Variance (σ^2 or s^2)

Variance measures the average squared distance of each individual data point from the arithmetic mean.

$$s^2 = \frac{1}{N-1} \times \sum_{i=1}^N (x_i - \bar{x})^2$$

b) Standard Deviation (σ or s)

The standard deviation is the square root of the variance, restoring the variability measurement back to the original units of the data.

$$s = \sqrt{s^2}$$

c) Range and Interquartile Range (IQR)

Difference between maximum and minimum values ($x_{max} - x_{min}$). Highly sensitive to extreme outliers. Range covered by the middle 50% of the ordered data ($Q_3 - Q_1$).

$$IQR = (Q_3 - Q_1)$$

1.1.3 Measures of Shape and Distribution Properties

The shape of a probability or frequency distribution describes how data points cluster across its range.

a) Skewness (Asymmetry)

Skewness measures the degree of asymmetry of a distribution around its mean.

$$Skewness = \frac{\frac{1}{N} \times \sum_{i=1}^N (x_i - \bar{x})^2}{s^3}$$

i. Positive Skew (>0)

Long tail extends to the right (mean $>$ median)

ii. Symmetrical ($=0$)

Mean and median are equal (Bell curve).

iii. Negative Skew (<0)

Long tail extends to the left (mean $<$ median).

b) Kurtosis (Tailedness and Peakedness)

Kurtosis measures the concentration of data in the tails relative to a normal distribution.

i. Leptokurtic (> 3)

Heavy tails and sharp peak; indicates frequent presence of extreme outliers.

ii. Mesokurtic ($= 3$)

Standard normal distribution tails.

iii. Platykurtic (< 3)

Light tails and a flatter peak.

1.1.4 Bivariate and Multivariate Association Measures

When examining relationships between two or more quantitative variables, statistical association metrics measure directional alignment and linear strength.

a) Pearson Correlation Coefficient (r)

Measures the strength and direction of a linear relationship between two continuous variables (X and Y).

$$r = \frac{\sum (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Range of $-1.0 \leq r \leq +1.0$

- i. $r = +1.0$: Perfect positive linear correlation.
- ii. $r = 0.0$: No linear correlation.
- iii. $r = -1.0$: Perfect negative linear correlation.